

Akhmedkhan (Ahan) Shabanov

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Education

Simon Fraser University

Ph.D. in Computer Science

Research on 3D computer vision

Supervisor: **Prof. Andrea Tagliasacchi**

Vancouver, Canada

Jan. 2023 – Present

Yandex School of Data Analysis (top 3% of 4000 applicants)

Additional master's-level program in CS and ML offered by Yandex

Featured courses: Self-Driving Cars, Deep (Reinforcement) Learning, Computer Vision, NLP, Algorithms, Crowd Sourcing, Game Theory

Moscow, Russia

Sep. 2018 – Jun. 2020

Moscow Institute of Physics and Technology

M.S. in Informatics and Computer Technology (top 5%, Honour degree)

GPA: 4.8/5; Thesis: Monocular 3D head with hair reconstruction

Supervisor: **Dr. Dmitry Ulyanov**

Moscow, Russia

Sep. 2019 – Sep. 2021

B.S. in Applied Mathematics and Physics

GPA: 4.5/5; Thesis: Neural feature descriptors for multi-modal image registration

Moscow, Russia

Sep. 2015 – Aug. 2019

Research & Work Experience

Meta

Research Intern

Zurich, Switzerland

Sep. 2025 – Feb. 2026

- Led a research project on generative 3D reconstruction
- Developed a non-grid-aligned 3D Gaussian reconstruction method based on flow matching

Huawei

AI Research Intern

Vancouver, Canada

Jan. 2025 – July 2025

- Led research on controllable generative models as interactive game engines
- Built a prototype using video diffusion models with game engine-style conditioning
- Studied consistency and control in synthetic environments via simplified physics simulator

Avaturn (ex. In3D)

Computer Vision Research Engineer, Founding team member

Moscow, Russia

Mar. 2020 – Aug. 2022

Avaturn - startup building technology for 3D body/head scanning with a smartphone.

- Built core pipeline for single/multi-view 3D head and hair reconstruction
- Led research on depth denoising for iPhone TrueDepth sensor (published)
- Developed an automated animation retargeting pipeline for 3D human avatars

European Molecular Biology Laboratory (EMBL)

Deep Learning Research Intern

Heidelberg, Germany

Jul. – Sep. 2019, 2020

Advisor: Anna Kreshuk

- Analyzed cell nucleoli segmentation in time-lapse microscopy videos
- Developed self-supervised method for temporal consistency in segmentation (published)
- Adopted by EMBL researchers: significantly reduced annotation workload

Publications & Patents

Free-Range Gaussians: Non-Grid-Aligned Generative 3D Gaussian Reconstruction [ECCV 2026]

Ahan Shabanov, Peter Hedman, Ethan Weber, Zhengqin Li, Denis Rozumny, Gael Le Lan, Naina Dhingra, Lei Luo, Andrea Vedaldi, Christian Richardt, Andrea Tagliasacchi, Bo Zhu, Numair Khan

BANF: Band-limited Neural Fields for Levels of Detail Reconstruction [CVPR 2024]

Ahan Shabanov, Shrisudhan Govindarajan, Cody Reading, Lily Goli, Daniel Rebain, Kwang Moo Yi, Andrea Tagliasacchi

Lagrangian Hashing for Compressed Neural Field Representations [ECCV 2024]

Shrisudhan Govindarajan*, Zeno Sambugaro*, Ahan Shabanov, Towaki Takikawa, Weiwei Sun, Daniel Rebain, Nicola Conci, Kwang Moo Yi, Andrea Tagliasacchi

Methods and systems for generating a resolved three-dimensional (R3D) avatar [US Patent 2022]

Dmitry Ulyanov, Igor Pasechnik, Ahan Shabanov, Vadim Lebedev, Ilya Krotov, Nikolay Chinaev, Bulat Yakupov, Vsevolod Poletaev

Self-supervised Depth Denoising Using Lower- and Higher-quality RGB-D sensors [3DV 2020]

Ahan Shabanov, Ilya Krotov, Nikolay Chinaev, Vsevolod Poletaev, Sergei Kozlukov, Igor Pasechnik, Bulat Yakupov, Artsiom Sanakoyeu, Vadim Lebedev, Dmitry Ulyanov

Unsupervised temporal consistency improvement for microscopy video segmentation with Siamese networks [bioRxiv, 2021]

Ahan Shabanov, Daja Schichler, Constantin Pape, Sara Cuylen-Haering, Anna Kreshuk

Optical-to-SAR Image Registration Using a Combination of CNN Descriptors and Cross-Correlation Coefficient [International Conference on Machine Vision (ICMV) 2019, Amsterdam]

Ahan Shabanov, Sergey Gladilin, Evgeny Shvets

A multilayer neural network for determination of particle size distribution in Dynamic Light Scattering problem [Computer Research and Modeling Journal, 2019]

Ahan Shabanov, Mikhail Nikolaevich Petrov, Alexander Viktorovich Chikitkin

Teaching Experience

TA for 3D Computer Vision course

Created and graded assignments for graduate level course

Vancouver, Canada

Spring 2022/2023

TA for Machine Learning course

Conducted weekly tutorial sessions for third and fourth-year undergraduates

Moscow, Russia

Spring 2020

Personal/Notable Achievements

Increased State Academic Scholarship

Scholarship for excellent performance and research activity (top 10%, Dean's List)

2019, 2020, 2021

2nd place at Vision Hackathon

Hackathon on analyzing videos captured by cameras on driving cars.
Our team of five won 10k\$

Moscow, Russia

2018

2nd place at Junction Hackathon

Junction - biggest hackathon in Europe

Helsinki, Finland

2018

1st place in Physics at All-Russian Olympiad

It was hosted by MIPT. I was granted direct university admission eligibility nationwide

Moscow, Russia

2014

Skills

- **Programming:** Python, C, C++, Bash, Git, LaTeX
- **ML/CV:** Numpy, Pytorch, TensorFlow, Keras, OpenCV, Kornia
- **3D/Computer Graphics:** Blender, MeshLab, Polyscope, Pytorch3D, libigl, Open3D
- **Languages:** Russian(native), English(C1)